

Navigating by Difference

Ghost Autonomy and the field pursuing full autonomy:
a comparative strategy and guardrail framework

Comparators	Waymo · Baidu Apollo Go · Zoox · Mobileye · Tesla
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Audience	Investors and prospective OEM / Tier-1 / assessor partners
Standing	Strategy document; claims graded per the program's evidence ladder

HOW TO READ THIS DOCUMENT — THE CLAIM LADDER

Ghost Autonomy's working standard is that measured, proven, projected, and conjectured statements must be distinguishable on sight. This document applies that standard to itself and to every program it discusses. Statements about third parties are cited to public sources and dated; statements about Ghost carry one of five standings: **[MEASURED]** witnessed on hardware; **[PROVEN]** a mathematical result; **[DEMONSTRATED]** running software; **[PROJECTED]** an engineering estimate awaiting hardware; **[NOTIONAL]** a research direction, quotable as intent only. Relocating a claim down this ladder is treated here as method, not weakness.

Prepared as a standalone companion to the Ghost Autonomy technical corpus (Ten Articles, May 2026; EPU–Ghost Autonomy Documentation v3.0, June 2026; Governed Knowledge Systems trilogy, July 2026). Third-party figures are company-reported unless independently audited; sources are listed in the References.

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Executive Summary

No operator of autonomous vehicles today claims Level 5. The five most consequential programs in the field — Waymo, Baidu Apollo Go, Zoox, Mobileye, and Tesla — are, in their engineering substance, five different answers to the same question: *how does a fallible learned system earn the right to act without a human?* Waymo answers with saturation validation inside expanding geographic envelopes; Apollo Go with vehicle-cost collapse and state-coordinated deployment; Zoox with a purpose-built vehicle and a staged, exemption-based regulatory path; Mobileye with a formal safety envelope (RSS) and a validation-economics argument (True Redundancy) sold through the automotive supply chain; Tesla with a statistical wager on fleet-scale data and end-to-end learning. Each answer is partial, and each partiality is visible in the 2025–2026 record: Waymo suspended its entire freeway service after construction-zone failures; Apollo Go's Wuhan fleet froze en masse in a single correlated event; Zoox cannot charge a fare until a federal exemption issues; Mobileye's formal envelope lives in software above the very stacks it must police; Tesla operates 59 driverless vehicles against a decade of full-autonomy rhetoric and a court finding that its flagship brand name is “unambiguously false.”

Ghost Autonomy is not a sixth robotaxi program, and this document does not position it as one. Ghost is a pre-silicon research program with a specific architectural thesis: that the decisive layer of the autonomy problem is not perception or planning but *enforcement and accountability* — a risk-monotone actuation boundary terminating in a non-programmable analog veto [MEASURED] (a ~32 ns intervention latency witnessed on FPGA, the program's only measured hardware number), an admissibility-before-fusion evidence discipline, and a constitutionally governed memory in which every change to operating truth is a logged, challengeable event. The honest inventory in Section 4 states plainly what exists: one measured latency, a small set of proved theorems, a disciplined 400-page corpus, a self-published coverage map with eleven acknowledged gaps, and no silicon, no vehicle data, and no external validation. The program's differentiator is that it says so — its claim ladder is itself the product ethic.

The method of this document is to use the five programs' *differences* from Ghost — and from one another — as guardrails: constraints that bound what Ghost should attempt, claim, and build next. Fifteen guardrails are derived in Section 6. Three carry the most weight. First, *no Level-5 language*: the operating field treats L5 as an asymptote, the credible leaders refuse the term, and the program's residual cross-domain L5 rhetoric is repositioned here, in writing, as a research horizon rather than a claim. Second, *do not compete where capital decides*: Waymo has raised roughly \$27B, GM burned \$~10B before killing Cruise, and Apollo Go's advantage rides on a sub-\$30,000 vehicle; Ghost's viable terrain is the layer those balance sheets have not built — enforcement hardware and safety-case instrumentation priced for integration, not fleets. Third, *differentiate against Mobileye or not at all*: RSS is the incumbent formal-safety franchise, standardized through IEEE 2846 and shipped in tens of millions of chips; Ghost's claim must therefore be stated with precision — enforcement below software in non-programmable hardware, one-sided (risk-conservative) epistemics rather than two-sided rule envelopes, and memory-integrated accountability that RSS does not attempt.

Section 7 converts the guardrails into a navigable direction: a three-stage posture that sells the *safety instrument* first (conformance gates, claim ledgers, replay harnesses — artifacts the 2026 regulatory turn toward structured safety cases now rewards), embodies the boundary in an FPGA guard ECU second, and defers custom silicon until the program's own Ledger-B proof debts (foremost the one-sided trusted-scalar bound and the liveness frontier) are paid. The register of risks in Section 8 is deliberately unvarnished — including the capital asymmetry, the single-researcher evidence base, the open transfer question from the program's gasification provenance to automotive perception, and a brand-collision hazard with the defunct Ghost Autonomy Inc. (Mountain View, ~\$220M raised, shut down April 2024) that any diligence search will surface.

The conclusion is one sentence long: *in a field where every operator is discovering that scaling autonomy means scaling the ability to refuse, remember, and be audited, Ghost's rational strategy is to become the layer that does exactly that — and nothing else.*

1 Purpose and Method: Differences as Guardrails

1.1 Why this document exists

Ghost Autonomy's technical corpus is unusually explicit about what the program believes; it has been less explicit about where the program is *going*. The founding essays claim a substrate for autonomy across many domains; the safety corpus claims a provably bounded Level-4 driving architecture; the newest work (July 2026) generalizes the memory doctrine into a domain-neutral constitutional economy of recall. Each is internally coherent. Together they leave the direction underdetermined — a program that could be read as a chip company, a safety-certification methodology, a memory-systems venture, or an autonomous-driving stack, depending on which document one opens first.

This document resolves that underdetermination by triangulation rather than introspection. It takes the five most consequential external programs pursuing high automation — Waymo, Baidu Apollo Go, Zoox, Mobileye, and Tesla — and treats each one's *difference* from Ghost as a navigational constraint. A difference can bound Ghost in one of three ways: it can mark territory Ghost must not enter (because the incumbent's balance sheet or fleet decides the contest), territory Ghost must answer (because the incumbent has already solved a problem Ghost claims to solve, in a different way), or territory the incumbent has left vacant (because its architecture, economics, or epistemics cannot reach it). The output is a set of fifteen guardrails (Section 6) and a staged direction that survives all of them simultaneously (Section 7).

1.2 Why these five programs

The comparison set was chosen on one criterion: programs whose present operations or supply position make them the live definition of “pursuing full autonomy” in mid-2026. Waymo is the operating leader in paid driverless service; Apollo Go is the largest robotaxi operation by cumulative rides and the global cost benchmark; Zoox is the purpose-built-vehicle pole and the test of the United States exemption pathway; Mobileye is the supply-chain incumbent and — decisively for Ghost — the only comparator whose differentiation, like Ghost's, is a *formal safety model*; Tesla is the vision-only, end-to-end pole and the field's cautionary study in claim governance. A sixth program, GM's Cruise, appears throughout as the control case: approximately \$10B invested, a single dragging incident in October 2023, permit suspension, and corporate shutdown by December 2024 [15] — proof that in this field the failure mode of overreach is not underperformance but extinction.

Ghost differs from all six in kind, not merely in degree: it is pre-silicon, pre-fleet, pre-revenue, and — by its own published standard — pre-validation. The comparison is therefore not competitive benchmarking; it is the use of operating reality as an instrument panel for a program whose own instruments are still theoretical.

1.3 Sources and epistemic rules

Third-party facts are drawn from company disclosures, regulatory filings, and major-outlet reporting through July 16, 2026, and are cited inline; where a figure is company-reported and unaudited (most fleet and safety statistics in this industry), the text says so. Ghost-internal statements are grounded in the program's corpus — the Ten Articles consolidation (May 2026), the EPU–Ghost Autonomy Documentation v3.0 (June 2026), the Provably Bounded compendium, the Temporal State Management monograph, the program coverage map (February 2026, revised), the CEO Research Opportunity Assessment (July 9, 2026), and the Governed Knowledge Systems trilogy (July 2026) — and carry the claim-ladder standings defined on the title page. No statement in this document upgrades a Ghost claim above the standing the corpus itself assigns.

2 The Level-5 Question in 2026

2.1 What the taxonomy actually says

Under SAE J3016, the difference between Level 4 and Level 5 is a single variable: the operational design domain. Both levels require the automated driving system to perform the entire dynamic driving task and its own fallback with no expectation of human intervention; Level 4 does so within a bounded ODD (geography, road class, weather, speed), while Level 5 is “unconditional” — an unlimited ODD, driving anywhere a typically skilled human could [1]. Two caveats in the standard matter for any investor reading this field. First, J3016 is a descriptive taxonomy, not a safety standard: it “imposes no requirements” and confers no judgment of performance — a level claim is a scope claim, never a competence claim [2]. Second, levels attach to features, not vehicles, which is why regulators and researchers treat marketing constructions such as “Level 2+” — and undated promises of Level 5 — as category errors [3].

2.2 What the field actually claims

The empirical finding of this review is stark: **no operator with a driverless vehicle on a public road claims Level 5, and the credible leadership explicitly repudiates it.** Waymo’s co-CEO Dmitri Dolgov defines L5 as “anywhere, in any condition... no operational limitations” and states that Waymo is not pursuing it — the binding constraint being validation, not capability [32]; his predecessor John Krafcik said years earlier that autonomy “will always have constraints” [33]. Mobileye’s Amnon Shashua goes further and proposes retiring SAE levels altogether in favor of a product taxonomy — hands-on/off, eyes-on/off, driver/no-driver — arguing the L4/L5 distinction has lost practical meaning; his 2026 framing is “eyes-off to mind-off through the 2030s,” an asymptote, not a milestone [62, 57]. Baidu frames Apollo Go strictly as L4 within geofenced ODDs; Chinese regulation currently reaches only to L3 permits for consumer cars (granted December 2025, to Changan and BAIC — not Baidu) and city-licensed L4 pilots, with no L5 pathway in the national framework [9, 10]. Zoox describes an SAE L4 system in dense-urban ODDs and its CEO explicitly declines ubiquity predictions [54]. The single prominent claimant of imminent generalized autonomy is Tesla — and in December 2025 a California administrative law judge found its “Full Self-Driving” branding “unambiguously false,” forcing a rebrand and a subscription-only product while the company sues to reverse the ruling [81]. The pattern admits one reading: *Level 5 is not a race anyone is running; it is the vanishing point that orients five different Level-4 strategies.*

2.3 What regulation now rewards

The regulatory turn of 2025–2026, on both sides of the Pacific, is toward *structured safety cases* rather than either laissez-faire self-certification or prescriptive hardware rules. In the United States: NHTSA’s April 2025 AV Framework streamlined crash reporting and opened Part 555 exemptions to domestically built AVs; the agency granted Zoox the first-ever demonstration exemption in August 2025; the AV STEP transparency program was withdrawn in June 2026 in favor of exemptions plus FMVSS modernization (a same-day proposal began adapting brake-pedal standards to vehicles without manual controls) [4, 51, 5]. California’s April 2026 rules require a *structured safety case* and phased permits with mileage gates; Texas requires state permits as of September 2025; the House SELF DRIVE Act of 2026 — passed out of committee in February — would institutionalize safety-case filings nationally [6, 7, 8]. China’s MIIT draft rules (February 2026, mandatory July 2027) impose data-recorder “black box” and remote-assistance requirements on L3/L4 systems [10]. The validation state of the art beneath these regimes is scenario-based and case-based: ISO 21448 (SOTIF) requires shrinking the unknown-unsafe scenario space; UL 4600 requires a claims-argument-evidence safety case across the lifecycle; ISO/PAS 8800 (December 2024) extends this to AI components with runtime monitoring; IEEE 2846 — seeded by Mobileye’s RSS — standardizes the assumptions a planner may make about other road users [11, 12, 13, 14].

WHY THIS SECTION IS LOAD-BEARING FOR GHOST

Every element of the 2026 regulatory settlement — structured safety cases, black-box data recorders, runtime monitoring of AI components, explicit assumption registries — is an *accountability artifact*. That is the layer Ghost’s architecture is about: a conformance gate that fails closed, a claim ledger with graded standings, an append-only decision memory, an enforcement boundary that does not trust the intelligence above it. The market’s direction of travel is toward what Ghost builds; the field’s leaders are being pushed there by incident, recall, and statute. Ghost’s opportunity is to arrive from the other side — by construction rather than by remediation. That opportunity is real, and Section 4 states with equal clarity how little of it is yet reduced to evidence.

3 The Field in Profile, July 2026

Each profile below compresses to four registers: what the program is, where it stands (with dated figures), how it argues its safety, and its posture toward full autonomy — closing with the design-space pole it defines for Ghost’s navigation. Fleet and safety figures are company-reported unless noted.

3.1 Waymo — validation saturation inside expanding envelopes

What it is. Alphabet’s autonomous-driving unit; the operating leader in paid driverless service. The sixth-generation Waymo Driver carries 13 cameras, 4 lidar, 6 radar and external microphones — a deliberately redundant multimodal suite, cost-reduced roughly 42% against the prior generation and in rider-only service since February 2026 [17, 18]. The AI stack is hybrid by design: a large “Waymo Foundation Model” teacher (with a Gemini-derived driving VLM) distilled into onboard student models, retaining structured representations, HD-map inputs, and — disclosed December 2025 — a separate onboard trajectory-validation guardrail layer that bounds what the learned planner may do [19]. Waymo publishes end-to-end research (EMMA) but does not ship it as the driver [20].

Where it stands. Roughly 500,000 paid rides per week across 10–14 US metros (Phoenix, San Francisco, Los Angeles, Austin, Atlanta, Miami, Dallas, Houston, San Antonio, Orlando; driverless operations opening in Denver, Las Vegas, San Diego, Tampa as of July 8, 2026); about 3,000 vehicles; 220M+ rider-only miles through March 2026, accumulating at >4M miles/week; testing in Tokyo and London [21, 22, 23, 25]. Capital: a \$16B round closed February 2026 at a \$126B post-money valuation; approximately \$27B raised to date; third-party analysts still estimate multibillion-dollar annual losses [26].

How it argues safety. The most institutionalized safety case in the field: a public safety-impact hub benchmarked against human baselines (94% fewer serious-injury-or-fatal crashes, 82% fewer airbag crashes, through 220M miles), peer-reviewed crash-typology studies, large-scale scenario simulation, and the disclosed guardrail layer [23, 24]. The 2025–2026 record also shows the limits: a 1,212-vehicle recall (May 2025); a school-bus-passing defect that produced an NHTSA probe, an NTSB investigation, and a software recall (December 2025); and — most instructive — the suspension of *all* freeway service in May 2026 after construction-zone failures, followed by a ~3,900-vehicle recall [28, 29, 30].

Posture toward L5. Explicit refusal. Strategy is scaled L4: expand the ODD city by city, foundation models to cut per-city adaptation cost, a stated 2026 target of one million weekly rides [32, 31].

THE POLE IT DEFINES *Depth before breadth — safety as an institutional artifact, bought with capital and miles that no entrant can replicate. What it leaves open: even the deepest validation regime discovers its unknown-unsafe scenarios in production (school buses, construction zones), one software release at a time.*

3.2 Baidu Apollo Go — the cost curve and the correlated fleet

What it is. The world's largest robotaxi operation by cumulative rides, built on the RT6 — a purpose-built L4 vehicle with 38 sensors including 8 lidars, at a unit cost under \$30,000, roughly an order of magnitude below early Western robotaxi builds — and the Apollo ADFM foundation model, which Baidu describes as the first large model supporting L4 and claims (unaudited) to be “10× safer than human driving” [34, 35, 36].

Where it stands. 22M+ cumulative public rides by April 2026; 27 cities; 3.2M fully driverless rides in Q1 2026 alone (+120% YoY); weekly volume passed 350,000 in March 2026; over 1,000 fully driverless vehicles; unit-economics breakeven claimed in Wuhan since late 2024 (company-reported, segment financials undisclosed) [36, 37]. International expansion is asset-light and fast: Hong Kong, Dubai (with Uber and the RTA), Abu Dhabi, Seoul, a Swiss Level-4 permit across three cantons (June 2026, safety operators aboard), and London testing with Uber and Lyft targeted for 2026 [38, 39].

How it argues safety. Aggregate exposure claims (one airbag-deployment accident per ~12M driverless km) and an ASIL-D certificate for a software subsystem — but no published third-party methodology; Bloomberg's January 2026 assessment of the industry applies with force here: nobody outside can yet verify the safety multiple [40, 16]. The defining 2026 event is negative and systemic: on March 31, dozens of Wuhan robotaxis froze simultaneously after a platform-level malfunction — rear-end collisions, passengers trapped for up to two hours, SOS buttons failing, citywide suspension — with no full technical account published since [41].

Posture toward L5. None claimed. Robin Li's long-held thesis is that L4 commercializes before L3 because liability is cleaner; “L5” appears only around a 2021 concept car, never the operating program [42].

THE POLE IT DEFINES *Scale through cost collapse and jurisdictional pragmatism. What it leaves open: correlated, fleet-scale failure — the Wuhan stall is the first mass demonstration that in centralized fleets the unit of failure is not the vehicle but the platform, and no operator has published an architecture for fleet-level fail-safety.*

3.3 Zoox — the purpose-built vehicle and the exemption path

What it is. Amazon's robotaxi subsidiary (acquired 2020, ~\$1.2–1.3B): a ground-up bidirectional vehicle — no steering wheel, pedals, or driver seat — with ~40 sensors in four corner pods, each sweeping 270°, plus thermal cameras and microphones; NVIDIA-based onboard compute; carriage seating for four [43, 44].

Where it stands. Public rides in Las Vegas (open access since September 2025) and San Francisco (waitlisted since November 2025); Austin and Miami announced for late 2026; testing in seven-plus metros; ~2M driverless miles and 500,000+ riders claimed by mid-2026. Every ride is free: Zoox cannot lawfully charge until NHTSA grants its Part 555 commercial exemption (petition pending, comment period closed June 2026) — the first test of scaling a non-FMVSS-conformant vehicle design in the US. Its Hayward plant entered start-of-production in July 2026 with claimed capacity ramping toward 10,000+ units/year [45, 46, 47, 48, 49].

How it argues safety. A published quantitative safety-case framework (Safety Report v3.0: staged deployment tiers, simulation and closed-course verification, residual-risk clearance) — the closest any operator comes to a UL 4600-style public case — plus the historic August 2025 AVEP demonstration exemption. The record also carries three software recalls in 2025 (unexpected braking; an e-scooter contact; lane-crossing behaviors) and a 2024 NHTSA finding of apparent noncompliance with eight FMVSS provisions, resolved through the exemption route [50, 51, 52, 53].

Posture toward L5. Rejected in favor of dense-urban L4 with a rider-optimized vehicle. CEO Aicha Evans: “as fast as we can, but as slow as necessary”; mainstream ubiquity “maybe a decade from now, or maybe two — I'm not in the prediction business” [54].

THE POLE IT DEFINES *Vertical integration of the vehicle itself — and the discovery that the binding constraint on a novel architecture is regulatory throughput, not engineering. What it leaves open: purpose-built integration took a decade and Amazon's balance sheet; nothing about its safety argument is available as a component to anyone else.*

3.4 Mobileye — the formal-safety incumbent (Ghost’s true comparator)

What it is. The automotive supply chain’s autonomy incumbent: 36.7M EyeQ chips shipped in 2025 alone; a product ladder from base ADAS through SuperVision (hands-off), Chauffeur (eyes-off), and Drive (driverless); Volkswagen Group as anchor customer; a \$24.5B claimed design-win pipeline [55, 58, 57]. Two architectural theses define it. *True Redundancy*: run a camera-only subsystem and a radar/lidar subsystem as independent full perception channels, multiplying mean-time-between-failure and cutting the validation miles each channel must individually demonstrate [59]. *RSS* (Responsibility-Sensitive Safety): a published formal model of duty-of-care driving — safe distances, right-of-way, evasive obligations — wrapped around the learned driving policy so the planner is provably never the cause of a collision under stated assumptions; RSS seeded IEEE 2846 [60, 14]. Its 2025–2026 architectural language — Compound AI Systems versus end-to-end monoliths, a “fast/slow thinking” split in which a vision-language model advises but is barred from the control path — is a public argument that unconstrained end-to-end learning is unvalidatable [61].

Where it stands. FY2025 revenue \$1.89B (+15%); Q1 2026 \$558M (+27%) with raised guidance — against a \$3.79B goodwill impairment reflecting the gap between autonomy’s promise and its market pricing; Intel retains ~97% voting control [56]. Robotaxi commercialization is now direct: VW ID. Buzz vehicles with Mobileye Drive testing in Los Angeles with Uber (April 2026; commercial launch targeted late 2026), a Lyft/Marubeni Dallas program, and — announced June 16, 2026 — Mobileye’s own vertically integrated robotaxi business: ~100 driverless vehicles in a major US metro from 2027, scaling toward ~17,000 in five years [63, 64, 65]. The losses are also instructive: Zeekr — SuperVision’s flagship, 110,000+ vehicles — is migrating to an in-house NVIDIA-based stack, and Croatia’s Verne switched its robotaxi platform to Pony.AI [66, 67].

How it argues safety. The only comparator whose safety argument is *a priori* rather than purely statistical: a formal envelope (RSS) plus an independence argument (True Redundancy) that explicitly attacks the validation-mileage problem — tens of thousands of hours per channel rather than hundreds of millions for a fused stack (company claim) [59]. No major AV-defect recall or federal probe surfaced through mid-2026.

Posture toward L5. Shashua treats geographic-universal L5 as an asymptote and the SAE ladder as obsolete; the volume product is eyes-off consumer autonomy through the 2030s [62, 57].

THE POLE IT DEFINES *Formal safety as a business, delivered through the supply chain at consumer-electronics economics. What it leaves open — and this defines Ghost’s differentiation obligation: RSS is enforced in software, above the operating system, inside the same programmable stack it polices; its assumptions are two-sided estimates rather than one-sided bounds; and it maintains no governed memory — no constitutional account of how the system’s operating truth changes over time.*

3.5 Tesla — the statistical wager and the cost of claims

What it is. The vision-only, end-to-end pole: 8 cameras, no lidar, no radar, no HD maps; FSD v14 as the consumer build with a ~10× parameter-count jump (October 2025) and a v15 “architectural overhaul” promised for late 2026; HW4/AI4 onboard compute with AI5 taped out at TSMC and Samsung (volume 2027); Dojo shut down in August 2025 as “an evolutionary dead end,” training returned to NVIDIA clusters; the two-seat Cybercab entered production April 2026 via FMVSS self-certification [68, 69, 70, 71, 72].

Where it stands. The gap between narrative and operation is the analytical core. Operated reality as of mid-2026: robotaxi service in Austin (whole-metro geofence, ~20 active driverless vehicles), Dallas, Houston, and Miami with small geofences; Bloomberg counted **59 driverless vehicles fleet-wide** in June 2026; Bay Area service still uses human drivers under a limousine permit, with the CPUC stating Tesla “is not operating an autonomous vehicle service” in California; unsupervised consumer FSD has slipped to “probably Q4 2026” [73, 74, 75]. Narrative: a decade of imminent-L5 promises (coast-to-coast by 2017; a million robotaxis by 2020; L5 “this year” in 2020), a ~\$1T pay package gated on 1M robotaxis and 10M FSD subscriptions, and “unsupervised FSD widespread in the US by year-end” [76, 77].

How it argues safety. By fleet statistics — ~10B cumulative FSD (Supervised) miles by May 2026, 1.3M paying users — released in more granular form only after Waymo’s co-CEO publicly challenged the

methodology; long-standing criticisms (road-mix skew, airbag-only crash counting) persist, and Tesla's own data showed Autopilot safety regressing in 2025. Federal posture: an Engineering Analysis (recall-track) covering 3.2M vehicles for reduced-visibility FSD behavior; a 2.88M-vehicle probe over 58 traffic-violation incidents; an NTSB-joined investigation of a fatal June 2026 crash [78, 79, 80]. In December 2025 a California ALJ found the "Full Self-Driving" name "unambiguously false"; Tesla rebranded, moved FSD to subscription-only, and sued the DMV [81].

Posture toward L5. The only leadership claiming generalized autonomy imminently — and the only one whose claims a court has formally found false. SAE framing is avoided; the operated product is L2 (consumer) and small-geofence L4-style (robotaxi).

THE POLE IT DEFINES *Scale of data as the safety argument, and marketing as the enemy of the safety case. What it leaves open: an end-to-end stack with no published structural safety argument is precisely the architecture that cannot answer the 2026 regulatory turn — every accountability artifact regulators now demand is something this pole, by construction, does not produce.*

4 Ghost's Developing Technology: An Honest Inventory

This section states what Ghost has actually built, proved, measured, and merely proposed — at the resolution an acquirer's technical diligence would apply. It is deliberately the least promotional section of the document, because in this field (Section 3 passim) the promotional register is now a quantifiable liability.

4.1 The architectural thesis

Ghost's corpus develops one invariant and builds everything else around it: *as warrant for a benign reading of the scene falls, the set of admissible commands may only ever contract — and the final contraction is enforced by physics, not software.* The realization is a five-stage refusal chain. S0 treats numerics and sensing as safety variables (safe-side arithmetic; noise floors setting threshold floors). S1 is a fidelity gate built on a "trusted scalar" φ under a five-part contract, the decisive clause being one-sidedness: φ may err only toward *higher* risk. S2 is a layered safety filter in the control-barrier-function family (established literature, adopted rather than invented). S3 is a risk-monotone actuation algebra: admissible command sets are convex boxes indexed by risk band; risk up, box shrinks — never the reverse; composition is by intersection only; a rate clamp bleeds the envelope down over ~ 1.25 – 1.47 s rather than step-lurching the vehicle. S4 terminates the chain in an electrically isolated analog veto — hardwired comparators with no programmable logic in the guard path — that forces a minimal-risk condition when timing or plausibility contracts are violated. Above the chain sits an evidence discipline (*admissibility before fusion*: each sensor is a witness whose testimony must pass integrity, alignment, scope, and lineage gates before it may touch the fused world model; correlated witnesses count once; inadmissible evidence is excluded, never averaged in) and a constitutional governance layer (Archive, Proposers, Verifiers, Adjudicators; an Unknown Register; anti-silent-drift: every change to operating truth is a logged, challengeable event). The safety goal is restated as six acceptance criteria (G1–G6 — one-sided error, antitone admissibility, non-expansive enforcement, physical fail-closed, legible policy, conjunctive release), published explicitly as *criteria, not achievements*.

The intended contribution to the certification problem is structural rather than statistical: where ISO 21448 practice asks *how rarely* the stack errs, a conformance gate over G1–G6 asks *what an error can do* — converting part of the unbounded scenario-coverage problem into a bounded property-verification problem. Whether that conversion survives contact with real perception noise is, by the program's own ledger, an open obligation (Section 4.4).

4.2 The substrate program: G10, EPU, PICAPD

Beneath the safety architecture sits a proposed computing substrate with an unusual provenance. The founder's doctoral work (Illinois Institute of Technology) modeled biomass gasification — an aggressively nonlinear multiphase system — and found that complete elliptic integrals evaluated by the arithmetic-geometric mean (AGM) yield conservation-respecting reduced-order models with machine-precision convergence in a bounded, parameter-independent number of iterations [PROVEN], grounded empirically in the gasification corpus (200+ experimental datasets; pipeline $R^2 \approx 0.78\text{--}0.87$, reported as preliminary and valid only within the demonstrated operating envelope) [MEASURED]. The G10 “gate” methodology generalizes that result into a ten-stage constraint pipeline; the Event Processing Unit (EPU) is its proposed silicon embodiment (24 variational cores, an event-dispatch network, a sensor-fusion and state-projection unit); PICAPD is the accompanying instruction-set architecture, including event registers, constraint instructions, and a machine-safety control register. Headline figures attached to the EPU — ~ 3.4 ns decision latency, 10^6 decisions/s, 17–44 W typical draw, $\sim 23\times$ the power efficiency of a contemporary GPU — are engineering projections [PROJECTED]; the corpus's own provenance ledger marks the ISA-level performance tables as unverifiable at present and directs that they “must not masquerade as built.” There is no fabricated silicon. The single hardware-witnessed number in the entire program is the ~ 32 ns intervention latency of the analog veto on FPGA [MEASURED].

4.3 The memory module: governed knowledge systems

The July 2026 trilogy (*Symbiotic Search Fabric*; *The Retrieval Covenant*; *The Price of Recall*) extends the continuation-bet thesis — memory paired with constitutional protocol as the foundation of trustworthy autonomy — into a domain-neutral specification: retrieval as a priced, signed, refusable contract (deadline, quality floor, budget); storage strata ordered by residence time, with verification burden *increasing* in residence time (the “verification inversion”); eviction formalized as refusal in advance; a shadow-price controller governing recall economics; and an evolution pipeline in which recorded agent experience is the only legitimate warrant for design change. All three documents self-label as notional [NOTIONAL] — “nothing in this paper should be quoted as an achievement” — and no validation program has yet run. Strategically the trilogy matters for one reason: fleet learning is where every operator in Section 3 is weakest on accountability, and ISO/PAS 8800's runtime-monitoring and data-governance requirements pull directly toward governed memory. The trilogy is therefore inventoried here as a second product direction *subordinate* to the actuation-boundary wedge — consistent with the program's own July 2026 research-opportunity assessment, which ranks the hardware-enforced boundary first and pointedly declines to rank the trilogy at all.

4.4 What the program says about itself: coverage and proof debts

Ghost publishes a self-assessment most ventures would suppress. The program coverage map (February 2026, revised) grades 56 research subcategories across six domains — 15 FULL, 20 HIGH, 10 PARTIAL, 11 GAP — with the revision extending to 68 subcategories (21/28/9/10) and, critically, a ledger of *formal proof debts*: P-1, the one-sided trusted-scalar bound (graded CRITICAL — “without it the contract is aspirational”); P-2, the liveness frontier (a φ that always reports maximum risk is vacuously safe and parks the vehicle forever); P-3, non-box actuator geometry (friction-circle exactness); P-4, null-band calibration under real vibration and weather; through P-7, certification-boundary marking — a combined 90–131 person-days of estimated closure work. The acknowledged gaps cluster exactly where a paper program would be expected to be weakest: mechanical qualification (MEMS noise under vibration, ISO 16750-3 profiles, AEC-Q100 pathways) and far-horizon sensing; SOTIF integration, ADAS interfacing, and V2X are graded PARTIAL by the program itself.

4.5 Claim-standing summary

Standing	Contents of the inventory
[MEASURED]	~32 ns analog-veto intervention latency, witnessed on FPGA — the program’s only measured hardware figure. Gasification pipeline $R^2 \approx 0.78\text{--}0.87$ within its demonstrated envelope; dissertation model error 8.6% MAE against 15.7% for reference CFD, on 200+ datasets.
[PROVEN]	AGM evaluation of complete elliptic integrals: machine precision in bounded, parameter-independent iterations. Actuation algebra, box case: projection is coordinatewise clipping; firm non-expansion (1-Lipschitz); feasible-case exactness of the two-stage clamp.
[DEMONSTRATED]	Event-sourced multi-agent research laboratory with replayable checkpoints (partial: 24 of 53 specified components); browser-based demonstration suite (explicitly a visualization of the architecture, not evidence of hardware).
[PROJECTED]	All EPU silicon figures: ~3.4 ns decision path, 10^6 decisions/s, 17–44 W, ~23× GPU power efficiency, die cost and yield.
[PROPOSED]	The one-sided φ bound (P-1, CRITICAL); liveness frontier (P-2); non-box exactness (P-3); witness-object calibration (P-4); constitutional governance as engineering practice rather than architectural pattern.
[NOTIONAL]	The Governed Knowledge Systems trilogy; cross-domain constitutional parameterizations; any statement about domains beyond driving.

MATURITY, STATED WITHOUT CUSHION

Ghost today is a single-researcher program whose assets are a ~400-page disciplined corpus, one measured hardware number, a handful of theorems, a self-published gap ledger, and a claim-governance ethic. It has no silicon, no vehicle data, no pilot, no certification engagement, and no external validation of any result. The investment thesis cannot be that this is small — it is that it is *correctly shaped*: the field’s operating leaders are being driven by incident and statute toward exactly the artifacts this program has specified first, and capital converts a specification into the two proofs and one instrument that Section 7 defines as the near-term test of the whole thesis.

5 The Difference Matrix

Table 1 — set in landscape on page 12 — compresses the field onto the nine dimensions that matter for Ghost’s navigation. Three structural readings follow.

5.1 Reading 1: the field is converging on Ghost’s question — in software

Every mature program now interposes an explicit bounding layer between learned intelligence and actuation: Waymo disclosed its trajectory-validation guardrail in December 2025; Mobileye’s entire franchise is a formal envelope plus an architectural argument against end-to-end monoliths; Zoox’s staged tiers bound exposure procedurally; even Tesla is being walked toward constraint by engineering-analysis probes and a false-advertising ruling. This convergence is the strongest external validation of Ghost’s thesis — and the sharpest warning about its position. *The bounding layer in software is now contested, incumbent territory.* What remains vacant is the layer no operator has built: enforcement *below* the programmable stack — non-programmable, physically fail-closed, with one-sided rather than best-estimate epistemics — and the governed memory above it. Ghost’s differentiation must live in exactly those two strata, or nowhere.

5.2 Reading 2: validation economics is the binding constraint of the entire field

Waymo’s leadership names validation, not capability, as the reason L5 is not pursued; Mobileye’s True Redundancy exists to divide the required miles; Tesla’s wager is that ten billion supervised miles substitute for a case; Zoox’s constraint is a federal exemption queue. The industry’s scarcest commodity is not intelligence but *arguments that reduce required evidence*. Ghost’s structural claim — verify properties of the enforcement boundary instead of exhaustively sampling the behavior above it — is a third route between saturation (Waymo) and independence (Mobileye). It is also, by the program’s own ledger, unproven: P-1 and P-2 are precisely the theorems that decide whether the route exists. This is the correct interpretation of Ghost’s proof debts: they are not housekeeping; they are the investment thesis in mathematical form.

5.3 Reading 3: accountability is regulated into existence faster than it is engineered

China mandates ADS data recorders from July 2027; California now demands structured safety cases; ISO/PAS 8800 requires runtime monitoring and data governance for AI components; IEEE 2846 requires explicit assumption registries. Across the matrix row “memory & accountability,” no comparator offers a governed answer — Waymo’s is internal, Apollo Go’s is opaque (Wuhan’s technical account remains unpublished four months later), Tesla’s is adversarial, Mobileye’s stops at assumptions. The row is effectively empty at exactly the moment statute begins to demand it be filled. That vacancy — not the chip, not the gasification mathematics — is Ghost’s cleanest unclaimed ground, and it is addressable with software instruments years before custom silicon exists.

Table 1: Six programs, nine dimensions — July 2026. Ghost column carries claim-ladder standings; comparator cells are company-reported unless noted.

	Waymo	Apollo Go	Zoox	Mobileye	Tesla	Ghost (standing)
Primary artifact	Driverless ride service + Driver stack	Ride service + sub-\$30k purpose-built RT6	Purpose-built vehicle + service	Chips, stacks, formal model, via supply chain	Consumer feature (L2) + nascent robotaxi	Architecture, corpus, safety instrument [PROPOSED]
Sensing philosophy	Redundant multimodal (13 cam/4 lidar/6 radar)	Redundant multimodal (38 sensors, 8 lidar)	Redundant multimodal (~40 sensors, 270° pods)	Two independent channels: camera-only + radar/lidar	Vision-only, 8 cameras, no maps	Sensor-agnostic: admissibility gates on any suite [PROPOSED]
Where safety is enforced	Software guardrail layer over learned planner	Platform + in-vehicle redundancy (10 layers claimed)	Staged deployment tiers; software	RSS formal envelope — in software, above the stack	Inside the end-to-end network (implicit)	Below software: risk-monotone algebra ending in analog veto [MEASURED] (32 ns, FPGA)
Safety argument	Statistical at scale + scenario simulation + peer review	Aggregate exposure claims, unaudited	Published safety-case framework + exemption process	A priori formal model + channel independence	Fleet statistics, contested methodology	Structural: conformance gate over G1–G6; conversion of SO-TIF from frequency to consequence [PROPOSED]
Validation economics	Saturation: 220M+ miles, billions of sim miles	Exposure km + city permits	Simulation + closed course + staged tiers	True Redundancy: fewer miles per independent channel	10B+ supervised fleet miles as proxy	Property verification + twin-world replay; aims to shrink the miles problem, unproven [PROPOSED]
Memory & accountability	Internal; disclosed guardrail, no governed-memory claim	Undisclosed; post-incident opacity (Wuhan)	Internal; recall-responsive	Assumption registry via IEEE 2846; no memory governance	Data flywheel, no audit artifact; court-tested claims	Constitutional memory: append-only canon, anti-silent-drift, priced recall [NOTIONAL]
ODD posture	Expanding city envelopes; freeway service suspended May 2026	27 cities; geofenced; state-coordinated	Dense-urban, purpose-built ODD	Product ladder: hands-off → eyes-off → driverless	Small geofences operated; universal autonomy claimed	ODD-explicit by construction; refusal outside envelope is the product [PROPOSED]
Business model	Fleet operator (+ Uber channels); toward licensing (Toyota)	Fleet + asset-light international franchising	Amazon-funded fleet; free until federal exemption	Tiered supplier: \$50 chips to full stacks; now own robotaxi	\$99/mo subscription + future network	Layer supplier: instrument → guard ECU → substrate [PROPOSED]
Capital position	~\$27B raised; \$126B valuation (Feb 2026)	Baidu-funded; RT6 cost curve as capital substitute	Amazon (acq. ~\$1.2–1.3B, 2020) + ongoing	\$1.89B FY25 revenue; Intel-controlled	Self-funded from vehicle margins	Pre-seed; corpus-stage
2025–26 falsifier	Construction zones defeated freeway ODD; school-bus recall	Correlated fleet stall, Wuhan, Mar 2026	Cannot charge until Part 555 grant	Design-socket churn (Zeekr, Verne)	59 driverless vehicles vs. decade of L5 claims; DMV ruling	None yet run — the deficit and the point: P-1/P-2 are the first

6 Fifteen Guardrails, Derived from Difference

Each guardrail below is stated as a binding rule, followed by the evidence that makes it binding. Together they are the navigation instrument this document exists to produce: any proposed Ghost initiative — technical, commercial, or rhetorical — should be tested against all fifteen.

6.1 From Waymo: what validation depth teaches

G1 Claim no envelope you have not validated — and make standings visible on sight. Waymo, holding 220M+ rider-only miles and the field's deepest simulation apparatus, still suspended its entire freeway service when construction zones defeated it, and recalled software twice in seven months under federal scrutiny [29, 28]. If the operating leader's envelope can contract overnight, a pre-silicon program's claims must be graded, dated, and downgradable in public — which is precisely the claim-ladder discipline Ghost has adopted as method. The residual exception is dealt with at G13.

G2 Treat the safety case as a public, falsifiable artifact — the genre Waymo defined. Waymo's safety hub, peer-reviewed crash typologies, and disclosed guardrail layer established the evidentiary register in which credibility is now transacted [23, 24, 19]. Ghost's equivalents — the G1–G6 conformance gate, the proof ledger, replayable decision traces — must be published in refutable form, submitted where peer review can reach them, and never gated behind marketing. For a program whose product is an accountability layer, publication norms are product demonstrations.

G3 Do not compete with fleets on fleet terms. Waymo has raised ~\$27B and is valued at \$126B; GM spent ~\$10B before killing Cruise; both numbers price the cost of learning-by-operating [26, 15, 27]. No initiative of Ghost's may require owning vehicles, launching cities, or accumulating exposure miles. Ghost sells *into* the assurance stack of those who do — or into the supply chain that serves them.

6.2 From Apollo Go: what the cost curve and the correlated fleet teach

G4 Price the boundary for the RT6 cost curve. Apollo Go's decisive advantage is a purpose-built vehicle under \$30,000 — cost collapse as a capital substitute [34, 36]. An enforcement layer that adds meaningful cost to that bill of materials will not ship in the fleets that scale fastest. The guard-ECU embodiment must target a component-grade price point (lockstep MCU + analog path, not a data-center accelerator), and the EPU's projected efficiency figures matter commercially only insofar as they survive that constraint.

G5 Own the fleet-scale failure mode no operator has answered. Wuhan, March 31, 2026: dozens of robotaxis froze simultaneously on a platform-level fault — collisions, trapped passengers, failed SOS, citywide suspension, and no published technical account since [41]. The industry's minimal-risk condition is defined per vehicle; the demonstrated failure was per *fleet*. Ghost's architecture generalizes naturally — correlated-witness discounting, anti-silent-drift, constitutional quarantine of suspect updates — and should claim this problem by name: fleet-level MRC semantics, specified and simulated in the Stage- α instrument.

G6 Make auditability the counter-position to unaudited scale. Apollo Go's safety multiple, breakeven claims, and exposure kilometres are company-reported; independent verification does not currently exist in this industry [40, 16]. Ghost should build for the buyer who has noticed: signed witness logs, replayable canon, third-party-checkable conformance verdicts. The pitch is not “trust us instead”; it is “here is the instrument that makes trust unnecessary.”

6.3 From Zoox: what purpose-built integration and the exemption path teach

G7 Stay integrable; never become a vehicle program. Zoox needed a decade, Amazon's balance sheet, and a purpose-built factory to embody its architecture — and still enters new cities with retrofitted Highlanders [43, 49]. Ghost's boundary must remain a component: a stateless guard ECU with drive-by-wire authority and timestamped sensor taps, retrofittable to any platform, presuming nothing about the vehicle above it. Every design decision that couples Ghost to a particular vehicle architecture is a step toward a capital requirement it cannot meet.

G8 Speak the exemption-and-safety-case grammar natively. Zoox's path — FMVSS noncompliance findings resolved into the first AVEP demonstration exemption, now a pending Part 555 commercial petition — shows that United States deployment now runs through structured regulatory demonstration [53, 51, 48]. California's 2026 rules demand a structured safety case outright [6]. Ghost's conformance gate, claim ledger, and replay traces should be formatted from birth as safety-case exhibits — UL 4600 claims-argument-evidence structure, SOTIF-compatible scenario semantics — so that a partner's regulatory filing can incorporate Ghost artifacts verbatim [12, 11].

G9 Demonstrations must be evidence, not theater. Zoox validates through staged tiers, closed course, and simulation before public exposure [50]. Ghost's browser-based demonstration suite renders the architecture's story; by the program's own standard it evidences nothing. The next demonstration must produce measured quantities — the FPGA veto in a hardware-in-the-loop rig, fault-injection campaigns, latency distributions — or it should not be called a demonstration at all.

6.4 From Mobileye: what the formal-safety incumbent teaches

G10 Differentiate against RSS with surgical precision — or concede the ground. RSS is published, peer-visible, standardized through IEEE 2846, and shipping inside a \$1.9B-revenue supply position [60, 14, 55]. Ghost's corpus already cites it as foundation; the differentiation must therefore be stated exactly and defended nowhere else: (i) *locus* — RSS is enforced in software within the programmable stack it polices; Ghost's terminal enforcement is non-programmable and electrically isolated [MEASURED]; (ii) *epistemics* — RSS reasons from best-estimate assumptions about other agents; Ghost's trusted scalar is contractually one-sided, erring only toward risk [PROPOSED]; (iii) *scope* — RSS governs interaction rules; Ghost's chain governs the whole path from evidence admissibility to actuation, plus the memory of every change [NOTIONAL]. Any Ghost text that blurs these three deltas invites the correct rejoinder: "Mobileye shipped this years ago."

G11 Build a rung ladder — revenue at every stage of belief. Mobileye monetizes each step from \$50 ADAS chips to full driverless stacks, so every customer relationship funds the next rung [57]. Ghost's ladder is Section 7's staging: a *safety-instrument* product (software; sellable to safety organizations and assessors now), then the guard ECU (FPGA; sellable as a pilot component), then — only if economics demand it — the EPU substrate. A program that prices only the top rung sells nothing until everything is believed at once.

G12 Assume socket churn; make certification the moat. Zeekr — SuperVision's flagship, 110,000+ vehicles — is migrating to an in-house stack; Verne switched platforms outright [66, 67]. Technical merit does not hold automotive sockets; qualified evidence does. Ghost's durable asset should be assessor-facing: artifacts and relationships with the TÜV/UL/exida stratum and conformance to ISO 26262/21448/PAS 8800 grammars, which persist across OEM churn — the "assessor moat" the program's own opportunity assessment prescribes [13].

6.5 From Tesla: what claim governance teaches

G13 Never spend credibility ahead of evidence — retire the residual L5 language now. The controlled experiment has been run at civilization scale: a decade of imminent-autonomy claims terminated in a judicial finding that the product’s name itself was “unambiguously false,” a forced rebrand, recall-track federal probes — and an operated fleet of 59 driverless vehicles [81, 80, 73, 76]. Ghost’s founding essay still speaks of Level-5 autonomy across dozens of domains. This document formally relocates that language: cross-domain constitutional autonomy is a research horizon **[NOTIONAL]**; the program’s claim is provably *bounded* L4 enforcement, and no Ghost surface should carry an undated capability promise or an SAE level as destiny. In a program whose brand is claim discipline, one hyperbolic sentence costs more than a missing feature.

G14 Data volume is not a safety case — and neither is corpus volume. Ten billion supervised miles have not bought Tesla a structural argument; granular data arrived only under public challenge, and the methodology remains contested [78, 79]. The mirror-image trap for Ghost: four hundred disciplined pages are not proofs. The corpus earns its standing only when P-1 and P-2 close and when the measured-number count rises above one. Volume — of miles or of pages — must never appear in Ghost materials as evidence.

G15 Refuse the sensor war entirely. Vision-only versus lidar is a cost-philosophy dispute between balance sheets; every camp’s choice is defensible on its own economics [68, 17, 59]. Ghost’s admissibility layer is sensor-agnostic by construction — witnesses are gated by integrity and scope, whatever the modality. Taking a side would forfeit the only commercial position available to a layer supplier: serving all camps. Ghost documents should treat sensing choices the way a court treats witnesses — rules of evidence, no favorites.

6.6 Synthesis: the position the guardrails triangulate

Read together, the fifteen rules bound a single coherent identity. Ghost must not operate fleets (G3), build vehicles (G7), fight sensor wars (G15), or price itself out of the cost curve that wins deployment (G4). It must not claim beyond its evidence (G1, G13, G14) nor blur its deltas against the formal-safety incumbent (G10). It must publish falsifiably (G2, G9), speak the regulatory grammar natively (G8), sell rungs rather than a summit (G11), and cultivate the assessor stratum as its moat (G12). And it holds two pieces of ground the entire field has left vacant: enforcement below the programmable stack (G10) and governed, fleet-scale memory and failure semantics (G5, G6). The identity that survives every constraint simultaneously:

THE POSITION

Ghost Autonomy is the enforcement-and-accountability layer for bounded autonomy: a physically fail-closed actuation boundary, an evidence-admissibility discipline, and a constitutionally governed memory — delivered first as safety-case instrumentation, then as a component-grade guard ECU, and only thereafter as silicon — sold to the programs that drive, not competing with them.

7 Strategic Navigation: From Guardrails to a Direction

7.1 Three postures, evaluated

Full-stack AV developer — rejected. Requires the capital of G3's counterexamples (Waymo ~\$27B; Cruise ~\$10B, terminated) and the operational machinery of Section 3 entire. Ghost holds neither, and the posture would place it in direct evidentiary competition with programs holding eight orders of magnitude more driving data.

Pure IP licensing — rejected. The inverse temptation: publish, patent, and license the mathematics without embodiment. It fails on the field's core currency: nobody licenses an unproven safety property, because the licensee inherits the burden of proof (G14). The defunct Mountain View namesake (Section 8.5) is the cautionary instance of software-only autonomy claims outrunning embodiment. Some embodiment is mandatory; the question is how little suffices per stage.

Staged layer supplier — adopted. The rung ladder of G11, gated by falsifiable exits, embodying no more than each rung requires.

7.2 The three stages, with falsifiers

Stage	Deliverables	Falsifier / exit condition
α — Instrument (~2 quarters)	Close P-1 (one-sided φ bound) and P-2 (liveness frontier) — the two theorems the whole thesis rests on. Ship Conformance Instrument v0: the G1–G6 gate, claim ledger, and twin-world replay harness, exercised against a public automotive perception dataset and released for refutation. Open one assessor conversation with the artifacts as exhibits.	If P-1 is unprovable as stated, the φ contract is re-scoped in public — the ladder's own method. If the liveness frontier proves empirically vacuous (the boundary parks every vehicle), the structural thesis fails visibly and cheaply, before hardware capital is spent.
β — Boundary (quarters 3–6)	FPGA guard ECU (lockstep safety MCU + analog veto path) in a hardware-in-the-loop rig; fault-injection campaigns; extend the program's measured set from one number to a measured <i>family</i> (latency distribution, missed-intervention bound, envelope-bleed profile). One retrofit pilot on a partner test track, instrumented as a UL 4600-style evidence pack. Begin the vibration/AEC-Q100 gap-closure the coverage map already prices at 55–80 person-days.	If the veto path cannot hold its timing contract under ISO 16750-3 vibration profiles, qualification precedes any scale claim — the map's own GAP cluster made a gate rather than a footnote. If no partner will host a track pilot on the Stage- α evidence, the market signal, not the mathematics, is the binding constraint.
γ — Substrate & memory (quarter 7+)	EPU silicon proceeds only if Stage- β economics show FPGA+MCU insufficient <i>and</i> rung revenue exists to fund it; every silicon figure remains [PROJECTED] until first measurement. In parallel, productize the memory module as fleet-learning governance (quarantine, priced recall, append-only canon) against ISO/PAS 8800's monitoring and data-governance pull — subordinate to, and credentialed by, the boundary wedge.	The substrate gate is explicitly reversible: if commodity safety silicon plus Ghost's architecture meets the contract, the EPU remains a research program and the company remains a layer supplier — a legitimate outcome, stated in advance.

7.3 Customers and counterparties, mapped to the field

The comparator set doubles as a customer taxonomy. Waymo-class integrators build assurance in-house and are near-term unreachable — they are the *benchmark*, not the buyer. The reachable strata are: mid-tier

operators and OEM autonomy programs that must answer the 2026 safety-case regimes without Waymo's headcount (via Tier-1s, per G12's socket logic); the assessor and standards stratum (TÜV/UL/exida; IEEE/ISO working groups), where Ghost's artifacts function as reference implementations; regulators seeking demonstration-grade instrumentation for exemption review (the AVEP/Part 555 grammar of G8); and — for the memory module specifically — any fleet operator whose OTA and fleet-learning practice will meet ISO/PAS 8800 and China's 2027 data-recorder mandates with nothing but logs. Apollo Go's cost curve defines the price envelope (G4); Tesla's litigation record defines the compliance pitch; Mobileye defines both the channel to imitate and the incumbent to displace at the margin where its software envelope cannot follow (below the stack, and into governed memory).

7.4 What is measured instead of rides

Ghost's dashboard, this stage of its life, must refuse the field's vanity metric. The program's key indicators are: theorems closed against the proof ledger (P-1 through P-7); the count of *measured* quantities (currently one); external refutations invited and survived; assessor engagements opened; pilot integrations instrumented; and claim-ladder violations on public surfaces (target: zero). A quarter in which the measured-number count rises from one to five is worth more than any announcement the field would notice.

8 Risk Register

Stated per the program's own standard: plainly, with mitigations that are actions rather than adjectives.

8.1 Capital asymmetry

The field's reference points are \$16B rounds and \$10B failures [26, 15]. *Mitigation:* the staged posture never requires fleet capital; Stage α is a proofs-and-software budget; Stage β is an FPGA-and-test-bench budget. The asymmetry is fatal only to postures this document rejects.

8.2 Single-researcher evidence base

The corpus is one person's work; the external-validation column is, at present, empty — the program's own assessment says so. *Mitigation:* Stage α is designed as an external-validation machine: public release for refutation, peer-review submission of the P-1/P-2 results, an advisory bench spanning functional safety and formal methods, and university partnership for the HIL campaigns. Every subsequent claim gains a second signature or it waits.

8.3 Methodology-transfer risk

The program's strongest empirical results live in biomass gasification; the corpus itself marks transfer to automotive perception as an open validation obligation [PROPOSED]. *Mitigation:* Stage α runs the instrument against automotive data from day one; the gasification provenance is presented as provenance — never as automotive evidence.

8.4 Residual claim-governance debt

Fragments of the pre-discipline register survive (the cross-domain Level-5 essay; legacy knowledge-base labels), and under G13 they are liabilities. *Mitigation:* this document performs the repositioning for the strategic layer; the July 2026 website update plans already schedule the surface sweep; the rule going forward is standings-on-sight, everywhere, including investor materials.

8.5 The name

“Ghost Autonomy” is also the name of a defunct Mountain View venture — ~\$220M raised from Sutter Hill, Founders Fund, Coatue and the OpenAI Startup Fund; camera-first highway autonomy promised for 2020 and never shipped; a late pivot to LLMs-for-driving publicly criticized as premature; shut down April 3, 2024 [82, 83]. Every diligence search will surface that history first, and its failure pattern — claims outrunning embodiment — is precisely the pattern this program’s ethic exists to negate. *Mitigation*: a deliberate naming decision before external fundraising (distinct legal name, qualifier, or rename), a trademark and domain search, and a one-paragraph disambiguation in the data room that converts the collision into a statement of contrast.

8.6 Regulatory drift

The thesis gains force from the 2026 turn toward structured safety cases; a durable federal swing toward bare self-certification with broad preemption (the House bill’s direction) would weaken the compliance pull [8, 5]. *Mitigation*: the instrument is priced as engineering value (validation-cost reduction, per Reading 2), not as compliance tax; statutes then become tailwind rather than premise. The exposure is monitored, not assumed away.

9 Conclusion

The five programs pursuing full autonomy have, among them, raised on the order of fifty billion dollars, built three purpose-built vehicles, shipped tens of millions of chips, and accumulated hundreds of millions of driverless miles — and not one of them claims Level 5, because not one of them has an argument that survives an unlimited domain. What they are all discovering, by recall, by suspension, by stall, by statute, and in one case by judicial finding, is that the scarce commodity in autonomy is not intelligence but *warranted restraint*: the engineered ability to refuse, to remember, and to be audited. That is the layer Ghost Autonomy has specified — with one measured number, a small set of theorems, a self-published ledger of its own debts, and a claim discipline the field’s leaders adopt only under subpoena. The strategy this document derives is correspondingly narrow and falsifiable: prove the two theorems everything rests on, ship the instrument that makes safety cases checkable, embody the boundary in hardware one rung at a time, and let the fleets drive. In a field navigating by ambition, Ghost’s compass is difference — and the direction it points is not toward the vanishing point the industry stopped chasing, but toward the layer every chaser now has to buy.

A Selected Verifiable Figures

Program	As of	Figure	Source
Waymo	Jul 2026	~500,000 paid rides/week; driverless ops opening in 4 further metros	[21]
Waymo	Mar 2026	220M+ rider-only miles; 94% fewer serious-injury-or-fatal crashes vs. human benchmark (company-published)	[23]
Waymo	Feb 2026	\$16B round at \$126B post-money; ~\$27B raised to date	[26]
Waymo	May–Jun 2026	All freeway service suspended; ~3,900-vehicle recall (construction zones)	[29, 30]
Apollo Go	Apr–May 2026	22M+ cumulative rides; 27 cities; 3.2M driverless rides in Q1 2026 (+120% YoY)	[36]
Apollo Go	2024–26	RT6 unit cost ~\$28–30k; Wuhan unit-economics breakeven claimed (unaudited)	[34, 36]
Apollo Go	Mar 31, 2026	Correlated fleet stall in Wuhan; collisions, trapped passengers, citywide suspension	[41]
Apollo Go	Jun 12, 2026	Swiss federal L4 permit (AmiGo, with PostBus), 80 km ² across three cantons	[38]
Zoox	Jul 6, 2026	Hayward plant start-of-production; claimed ramp toward 10,000+ units/yr	[49]
Zoox	Mid-2026	~2M driverless miles; 500,000+ riders; rides free pending Part 555 grant	[54, 48]
Zoox	Aug 6, 2025	First-ever NHTSA AVEP demonstration exemption	[51]
Mobileye	FY2025	36.7M EyeQ shipped; revenue \$1.894B (+15%)	[55]
Mobileye	Apr 23, 2026	Q1 revenue \$558M (+27%); guidance raised; \$3.79B goodwill impairment	[56]
Mobileye	Jun 16, 2026	Vertically integrated robotaxi business announced: ~100 vehicles in 2027 → ~17,000 in five years	[65]
Tesla	Jun 10, 2026	59 driverless robotaxis fleet-wide (Bloomberg count) across small geofences	[73]
Tesla	May 2026	~10B cumulative FSD (Supervised) miles; 1.3M paying subscribers	[78]
Tesla	Dec 2025–Feb 2026	“Full Self-Driving” found “unambiguously false” by CA ALJ; rebrand; Tesla sues DMV	[81]
Tesla	Mar 18, 2026	NHTSA Engineering Analysis EA26002 (recall track), 3.2M vehicles	[80]
Cruise (control)	Dec 2024	GM ends robotaxi funding after ~\$10B invested	[15]
Ghost Inc. (namesake)	Apr 3, 2024	Unrelated Mountain View venture; ~\$220M raised; shut down	[82]
Ghost (this program)	Jul 2026	One measured hardware figure (~32 ns FPGA veto); coverage self-graded 21/28/9/10 across 68 subcategories; proof debts P-1..P-7 open	corpus

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